

TUKTUT NOGAI, NT, Canada

WMO: 714920

Lat: 69.199N

Lon: 122.359W

Elev: 1810

StdP: 13.76

Time Zone: -7.00 (NAM)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-35.3	-32.2	-42.7	0.5	-35.2	-39.1	0.6	-32.1	47.2	2.8	43.2	0.5	7.1	250	1.065

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.5	67.8	55.5	64.4	54.2	61.0	52.5	57.5	64.6	55.3	62.6	53.0	59.9	12.4	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
54.0	66.4	60.6	51.4	60.3	58.5	49.2	55.5	56.1	25.6	65.0	24.1	62.6	22.7	60.4	65.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 39.7	34.4	29.4	DB	-39.9	74.6	3.8	2.4	-42.6	76.3	-44.8	77.7	-46.9	79.0	-49.6	80.7	(4)
(5)			WB	-40.4	60.5	3.7	2.6	-43.1	62.4	-45.2	63.9	-47.3	65.3	-50.0	67.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.0	-10.8	-10.9	-9.0	6.6	24.0	42.1	49.8	43.8	33.4	16.6	0.1	-7.2		
	DBStd	24.51	11.99	11.70	14.35	11.71	10.59	9.30	8.21	7.98	9.35	7.86	9.43	11.16		
	HDD50	12943	1884	1707	1828	1301	807	272	109	229	501	1034	1498	1773		
	HDD65	18243	2349	2127	2293	1751	1271	689	471	659	948	1499	1948	2238		
	CDD50	176	0	0	0	0	1	33	103	36	3	0	0	0		
	CDD65	2	0	0	0	0	0	0	1	1	0	0	0	0		
	CDH74	3	0	0	0	0	0	0	2	1	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	11.5	11.2	11.0	10.2	11.3	11.8	11.7	11.2	10.9	12.4	13.9	11.5	11.0		
	PrecAvg	7.9	0.4	0.3	0.4	0.3	0.4	0.7	1.3	1.5	1.0	0.7	0.5	0.4		
(15) Precipitation	PrecMax	10.3	1.1	0.9	1.2	1.0	1.1	1.5	2.2	3.0	2.1	1.5	1.1	0.9		
	PrecMin	5.6	0.1	0.0	0.1	0.0	0.1	0.2	0.6	0.7	0.4	0.2	0.2	0.1		
	PrecStd	1.3	0.2	0.2	0.2	0.2	0.2	0.3	0.4	0.6	0.4	0.4	0.3	0.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.5	18.4	25.8	41.8	58.3	70.1	72.4	70.5	58.3	43.2	23.2	21.7		
		MCWB	24.9	16.8	23.8	37.3	50.1	55.5	59.4	58.2	51.4	40.6	22.5	20.8		
	2%	DB	19.0	15.1	21.6	33.8	47.6	65.0	68.9	64.8	53.1	33.5	19.5	16.1		
		MCWB	17.5	13.9	20.0	30.8	40.5	53.1	56.2	55.2	47.9	32.5	18.8	15.4		
	5%	DB	14.0	11.3	18.1	28.6	42.2	60.5	66.1	60.3	49.8	29.8	15.9	12.6		
		MCWB	13.0	10.5	16.8	26.4	36.9	51.0	55.0	52.8	45.5	28.8	15.0	11.8		
	10%	DB	6.2	7.0	13.2	24.3	38.5	56.2	63.0	56.3	46.2	27.3	13.2	8.9		
		MCWB	5.2	6.1	12.1	22.6	34.3	48.2	53.7	50.1	43.3	26.3	12.5	8.2		
	0.4%	WB	24.8	17.0	24.1	37.6	49.3	56.9	61.0	59.8	52.4	40.5	22.9	20.3		
		MCDB	26.4	18.1	25.6	42.0	58.8	68.2	69.8	68.1	56.4	43.1	23.2	21.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	17.8	13.9	20.2	31.0	41.3	54.2	58.4	56.2	48.5	32.7	18.8	15.5		
		MCDB	18.9	14.9	21.5	33.6	46.4	63.4	65.9	62.1	52.6	33.5	19.3	16.2		
	5%	WB	13.0	10.4	16.9	26.3	37.3	51.7	56.5	53.2	45.9	29.1	15.1	11.8		
		MCDB	14.1	11.3	18.2	28.5	41.5	59.3	63.3	60.0	49.3	29.7	15.7	12.4		
	10%	WB	5.3	6.1	12.0	22.7	34.6	48.9	54.6	50.9	43.3	26.5	12.5	8.2		
		MCDB	6.2	6.9	13.1	24.3	38.1	55.7	61.5	55.5	45.9	27.1	13.2	8.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.5	13.2	14.3	15.2	13.7	13.2	14.5	11.6	9.1	8.6	10.4	11.7		
		MCDBR	17.0	15.9	15.2	15.5	14.9	18.3	17.4	15.9	12.9	9.2	12.0	14.1		
		MCWBR	16.2	15.3	14.2	13.1	10.5	10.0	8.9	8.5	8.5	8.8	11.7	13.6		
		MCDBR	17.1	16.0	15.1	14.3	14.0	17.1	16.2	14.7	12.6	8.8	12.0	14.0		
	5% WB	MCWBR	16.3	15.4	14.3	12.3	10.2	9.8	9.4	9.0	8.7	8.5	11.8	13.6		
(40) Clear-Sky Solar Irradiance	taub		0.148	0.200	0.236	0.283	0.319	0.319	0.333	0.315	0.278	0.213	0.145	N/A		
	taud		1.902	2.077	2.235	2.162	2.177	2.417	2.422	2.488	2.491	2.138	1.914	N/A		
	Ebn at Noon		22	188	258	272	273	279	270	263	244	175	17	0		
	Edh at Noon		1	13	22	33	37	30	29	24	18	11	1	0		
(44) All-Sky Solar Radiation	RadAvg		7	145	656	1447	1980	1947	1639	1054	545	163	21	0		
	RadStd		1	15	29	60	133	142	135	123	62	20	2	0		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (5 neighbors)		+1.01	+2.12	+2.77	N/A	N/A	N/A	-398	-387	N/A	N/A	

Nomenclature: See separate page